Information Systems Management

Case Study Frameworks Exam Pack

Unit-wise frameworks, exam cues, and case-style examples

How to use this PDF tonight

For any case answer, first identify the business problem, then select the framework that fits the problem. Write in this pattern: **case fact + framework point + implication + recommendation**. Do not only define a framework. Show how it explains the case and what the company should do next.

Universal Case Answer Template

1. **Context:** Name the company/sector and the information system issue.
2. **Problem:** State the pain point: cost, speed, customer experience, risk, decision quality, data silos, supply chain delay, cyber threat, legacy system, or new digital competition.
3. **Framework:** Apply the most relevant framework unit-wise. Use 3-5 points, not a long theory dump.
4. **Evidence from case:** Link each point to a fact from the case.
5. **Business impact:** Explain impact on efficiency, revenue, customer intimacy, compliance, resilience, or competitive advantage.
6. **Recommendation:** Suggest practical action: integrate systems, improve data governance, adopt cloud, strengthen controls, redesign process, use BI/AI, or change governance.

Quick Framework Selection Map

Case clue	Best framework	Use this angle
Company invests in IT for growth	Six strategic objectives of IS	Show business value: operations, new models, intimacy, decisions, advantage, survival.
Manual delays or cross-department work	Business process + BPM	Map current process, remove delays, redesign flow.
Departments not sharing data	Enterprise applications / ERP	Single database, integrated processes, standardization.
Competition, digital platform, industry disruption	Porter forces / value chain / network strategy	Explain how IS creates or defends competitive advantage.
Cloud, BYOD, infrastructure choice	IT infrastructure + TCO + competitive forces	Balance scalability, cost, flexibility, risk.
Large data, dashboards, prediction	BI infrastructure, OLAP, data mining	Convert data into decision support.
Cyber attack, privacy, outage	Security and control framework	Risk assessment, policy, controls, audit, DR/BCP.
Supply chain delays or inventory swings	SCM, bullwhip, push-pull	Match demand and supply, reduce inventory, improve visibility.
Customer churn or personalization	CRM operational/analytical	Use customer data for service, retention, targeting.
Online marketplace or app business	E-commerce features, business/revenue models	Identify platform, revenue logic, digital market changes.

AI, expertise, knowledge sharing	KM value chain / AI techniques	Capture, store, share, apply knowledge; choose ML/expert system etc.
Decision making at different levels	Simon model, MIS/DSS/ESS, BI environment	Match decision type to system support.
Building or changing a system	SDLC, prototyping, agile, outsourcing	Choose development method based on uncertainty, risk, speed.

1 Unit 1 and 2: Role of Information Systems in Global Business Today

Strategic Business Objectives of Information Systems

Core idea: Firms invest in information systems because IT is not only a cost center; it directly supports business strategy and performance.

Components / steps:

- Operational excellence: faster, cheaper, more reliable operations.
- New products, services, and business models: digital channels, platforms, apps, subscription services.
- Customer and supplier intimacy: better service, personalization, stronger supplier coordination.
- Improved decision making: real-time, accurate information instead of guesswork.
- Competitive advantage: lower cost, differentiation, better response, or unique capability.
- Survival: regulatory compliance, industry norms, and competitive necessity.

When to apply in a case: Use when a case asks why a company invested in an information system or what business value IT created.

How to write in the exam: Write: "The system supports the strategic objective of X because the case shows Y. This improves Z business outcome." Cover 3-4 objectives that fit the case.

Simple example: In a bank without branches like N26, mobile banking supports new business model, customer intimacy through app-based service, operational excellence through low branch cost, and competitive advantage against traditional banks.

Input-Processing-Output-Feedback Model

Core idea: An information system takes raw data, processes it, produces useful information, and uses feedback to improve decisions or operations.

Components / steps:

- Input: raw transaction/customer/operational data.
- Processing: sorting, calculating, classifying, summarizing, analyzing.
- Output: reports, dashboards, alerts, documents, decisions.
- Feedback: output returned to improve future input or processing.

When to apply in a case: Use when a case asks how an information system works or what data/reports should be created.

How to write in the exam: Identify the inputs, processing logic, outputs, and feedback loop. This is a strong structure for system design questions.

Simple example: For Astra customer-driven service, inputs are customer usage data, processing is analytics, outputs are customer segments/offers, and feedback is customer response that improves future targeting.

Dimensions of Information Systems

Core idea: A successful system is not just technology. It requires alignment between organization, management, and technology.

Components / steps:

- Organization: structure, processes, culture, politics, people, hierarchy.
- Management: decisions, strategy, coordination, control, change leadership.
- Technology: hardware, software, databases, networks, security tools.

When to apply in a case: Use when a case has implementation failure, user resistance, unclear ownership, or business-technology mismatch.

How to write in the exam: Explain the issue across all three dimensions, then recommend a balanced fix.

Simple example: If a hospital digitization project such as Jurong Health succeeds, it is not only because of software; clinical workflows, doctors, nurses, data standards, and management sponsorship must also align.

Business Information Value Chain

Core idea: Information systems create value only when data is transformed into useful decisions and business actions.

Components / steps:

- Raw data from transactions and environment.
- Information processing and storage.
- Management activities: planning, coordination, control, decision making.
- Business processes and firm performance.

When to apply in a case: Use when asked how data becomes business value.

How to write in the exam: Do not stop at "data was collected." Show how data supported decisions, process changes, and measurable results.

Simple example: In big data baseball, player data has value only when analytics changes scouting, training, game strategy, or salary decisions.

Sociotechnical Systems Approach

Core idea: Optimal performance comes from jointly improving the technical system and the social system. Technology alone does not guarantee productivity.

Components / steps:

- Technical side: systems, networks, databases, applications.
- Social side: people, roles, incentives, culture, training, routines.
- Joint optimization: design both together so users actually adopt the system.

When to apply in a case: Use in cases involving failed implementation, low adoption, organizational resistance, or change management.

How to write in the exam: Write that the firm must redesign processes, train users, change roles, and align incentives along with installing technology.

Simple example: For virtual education at Ahlia University, the platform matters, but faculty training, student engagement, assessment design, and support processes decide whether e-learning works.

2 Unit 3: Global E-Business and Collaboration

Business Process Framework

Core idea: A business is a collection of activities that produce a product or service. IT improves the flow of material, information, and knowledge.

Components / steps:

- Functional processes: sales, finance, HR, production.
- Cross-functional processes: order fulfillment, procurement, customer service.
- IT impact: automation, parallel work, reduced delays, new digital flow.

When to apply in a case: Use when a case describes operational delays, manual steps, departmental silos, or service redesign.

How to write in the exam: Map the process before and after IT. Mention which manual steps were automated and what delays were removed.

Simple example: City of Mississauga going digital can be explained as redesigning citizen-service processes so applications, approvals, and information flow online instead of through slow office visits.

Systems for Different Management Groups: TPS, MIS, DSS, ESS

Core idea: Different levels of management need different information systems because their decisions differ in routine, scope, and uncertainty.

Components / steps:

- TPS: records routine daily transactions for operational staff.
- MIS: summarizes TPS data into regular reports for middle managers.
- DSS: supports semi-structured decisions using models and analysis.
- ESS: supports senior executives with dashboards, trends, and external data.

When to apply in a case: Use when a case asks what system is needed for operational, middle, or senior management.

How to write in the exam: Match the decision level to the system. Give examples of reports or dashboards.

Simple example: In an e-commerce dashboard case, TPS records orders, MIS shows daily sales, DSS supports pricing or inventory analysis, and ESS shows strategic KPIs for top management.

Enterprise Application Architecture

Core idea: Enterprise applications integrate major business functions so information flows across the firm instead of staying in departmental silos.

Components / steps:

- ERP: integrates internal processes and common database.
- SCM: coordinates suppliers, production, inventory, logistics.
- CRM: manages sales, marketing, service, and customer data.
- KMS: captures and applies organizational knowledge.

When to apply in a case: Use when a case involves firm-wide integration, cross-functional data sharing, or a digital transformation program.

How to write in the exam: State which enterprise application fits the problem and explain its business role.

Simple example: Lenzing balancing supply and demand fits SCM; Adidas using CRM fits customer intimacy; Versum ERP transformation fits internal integration.

E-Business, E-Commerce, and E-Government

Core idea: Digital technologies can support internal business processes, online buying/selling, and government service delivery.

Components / steps:

- E-business: use of digital technology to run major business processes.
- E-commerce: buying and selling goods/services through the Internet.
- E-government: online delivery of government information and services.

When to apply in a case: Use when distinguishing digital business from online selling or public-sector digitization.

How to write in the exam: Define the category and connect it to the case actor: company, marketplace, university, or government.

Simple example: Mississauga is e-government; Deliveroo is e-commerce and mobile commerce; a firm using cloud tools internally is e-business.

Collaboration and Social Business Framework

Core idea: Collaboration systems improve productivity, quality, innovation, customer service, and financial performance when culture supports teamwork.

Components / steps:

- Drivers: changing work, professional knowledge work, innovation, globalization.
- Tools: email, IM, wikis, video meetings, cloud docs, enterprise social networks.
- Culture: from command-and-control to team-based, transparent collaboration.

When to apply in a case: Use when a case involves remote work, virtual education, digital teams, knowledge sharing, or social platforms.

How to write in the exam: Mention both tool and culture. A collaboration platform fails if the organization does not reward teamwork.

Simple example: Virtual education requires video platforms and LMS tools, but also collaboration norms between faculty, students, and administration.

IT Governance

Core idea: IT governance defines decision rights, accountability, and policies for using IT in the organization.

Components / steps:

- Who decides IT priorities?
- Centralized or decentralized IT function?
- Who owns data, security, budgets, and standards?
- How are business and IT aligned?

When to apply in a case: Use when a case asks about control, accountability, IT strategy, or digital transformation leadership.

How to write in the exam: Write about decision rights and accountability, not only technology.

Simple example: For a hospital digital transformation, governance clarifies who owns patient data standards, security, clinical workflow design, and project priorities.

3 Unit 4: Information Systems, Organizations, and Strategy

Organization: Technical and Behavioral Views

Core idea: Organizations can be viewed as formal structures processing inputs into outputs, and also as social systems shaped by culture, politics, routines, and conflict.

Components / steps:

- Technical view: inputs, production, outputs, legal entity, rules.
- Behavioral view: rights, obligations, routines, culture, politics, conflict resolution.
- IS impact: systems change both process efficiency and power/information relationships.

When to apply in a case: Use when a case asks how IT affects an organization or why people resist IT.

How to write in the exam: Explain both the formal process change and the human/political impact.

Simple example: A branchless bank changes the technical delivery model, but it also changes employee roles, customer trust, and the culture of banking.

Routines, Business Processes, and Culture

Core idea: Organizations depend on routines: standard ways of doing work. Business processes are collections of routines, and culture can either support or block change.

Components / steps:

- Routines: repeated standard operating procedures.
- Business processes: linked routines that produce outcomes.
- Culture: assumptions about what the organization values and how work should be done.
- Politics: groups resist systems that reduce their control over information.

When to apply in a case: Use for implementation resistance or process change cases.

How to write in the exam: Identify which routines are being changed and which cultural/political barriers exist.

Simple example: In a shipping wars case, a logistics platform changes routines around booking, tracking, pricing, and customer communication.

Transaction Cost Theory

Core idea: IT lowers the cost of participating in markets, making outsourcing and coordination with external partners easier.

Components / steps:

- Transaction costs include search, negotiation, monitoring, and coordination costs.
- Traditionally firms vertically integrated to reduce uncertainty.
- IT reduces market transaction costs, so firms can use external suppliers more efficiently.

When to apply in a case: Use when a case has outsourcing, platform coordination, supplier networks, or virtual firms.

How to write in the exam: Show how IT makes external coordination cheaper than doing everything internally.

Simple example: Li and Fung style virtual production works because IT coordinates suppliers without owning factories, materials, or ships.

Agency Theory

Core idea: As firms grow, supervision costs rise. IT reduces agency costs by making information visible and monitoring easier.

Components / steps:

- Agency costs: cost of managing and supervising employees/agents.
- IT enables dashboards, workflow tracking, automated reports, and performance transparency.
- Result: wider span of control and flatter organizations.

When to apply in a case: Use when IT enables monitoring, decentralization, or flatter structures.

How to write in the exam: Explain how better information reduces the need for layers of managers.

Simple example: A real-time delivery dashboard lets managers monitor drivers, orders, and delays without adding supervisors in every location.

Porter's Competitive Forces Model

Core idea: A firm must respond to five competitive forces. Information systems can weaken threats or strengthen the firm's position.

Components / steps:

- Rivalry among existing competitors.
- Threat of new entrants.
- Threat of substitute products/services.
- Bargaining power of customers.
- Bargaining power of suppliers.

When to apply in a case: Use when a case asks about strategy, competitive advantage, industry disruption, or digital platforms.

How to write in the exam: Take 3-5 forces and connect each to case evidence. Then show how IS helps.

Simple example: In Shipping Wars, digital logistics systems can reduce customer power through better service, reduce rivalry through tracking reliability, and improve supplier coordination.

Information System Strategies for Competitive Advantage

Core idea: IS can help firms deal with competitive forces through cost leadership, differentiation, focus on market niche, and stronger customer/supplier intimacy.

Components / steps:

- Low-cost leadership: use IT to reduce operating cost.
- Product/service differentiation: create unique digital service.
- Focus on market niche: analyze data for specific segments.
- Customer/supplier intimacy: lock in relationships through better integration.

When to apply in a case: Use when a question asks how IS contributes to strategic advantage.

How to write in the exam: Name the strategy and show the digital mechanism that supports it.

Simple example: N26 differentiates through branchless mobile banking; Grab improves customer intimacy using app data, payments, maps, and service personalization.

Value Chain Model

Core idea: The value chain sees the firm as a set of activities where value is added. IT should be applied at leverage points.

Components / steps:

- Primary activities: inbound logistics, operations, outbound logistics, sales/marketing, service.
- Support activities: procurement, technology, HR, firm infrastructure.
- Goal: identify where IT can reduce cost or create differentiation.

When to apply in a case: Use when analyzing where IT creates value inside a company.

How to write in the exam: Map the case to value-chain activities and identify the strongest IT intervention.

Simple example: For e-commerce delivery, IT adds value in order capture, route optimization, warehouse coordination, delivery tracking, and customer service.

Value Web

Core idea: A value web is a network of firms using IT to coordinate their value chains collectively.

Components / steps:

- Coordinates customers, suppliers, distributors, and partners.
- Flexible and adaptive to changes in demand and supply.
- Stronger than a single-firm value chain when the business depends on partners.

When to apply in a case: Use in platform, ecosystem, supply chain, smart city, or outsourcing cases.

How to write in the exam: Show how multiple actors exchange information through shared digital infrastructure.

Simple example: Singapore Smart Nation is closer to a value web because government agencies, citizens, businesses, sensors, payments, and public services are digitally connected.

Network-Based Strategies

Core idea: Network technologies create value because the usefulness of a product or service increases as more users join.

Components / steps:

- Network economics: more users create more value.
- Platforms attract complementary services and partners.
- Switching costs and ecosystem lock-in can increase.

When to apply in a case: Use when cases involve platforms such as Grab, Deliveroo, social networks, marketplaces, or payment systems.

How to write in the exam: Explain both sides of the network: users and providers/partners.

Simple example: Grab becomes more valuable when more riders, drivers, merchants, payment partners, and service categories join the platform.

Virtual Company Model

Core idea: A virtual company uses networks to link people, assets, and ideas without being limited by traditional boundaries or ownership.

Components / steps:

- Uses partners for manufacturing, logistics, design, or service delivery.
- Focuses on coordination rather than ownership.
- IT enables control, visibility, and collaboration across boundaries.

When to apply in a case: Use when a firm uses outsourcing, alliances, cloud services, or external partners.

How to write in the exam: Write that competitive advantage comes from orchestration, not owning every asset.

Simple example: A fashion brand using Li and Fung can design and sell products while partners manage sourcing, manufacturing, quality, and shipping.

Business Ecosystems and Platforms

Core idea: Many digital firms operate in ecosystems where one or two keystone firms provide platforms used by many niche firms.

Components / steps:

- Keystone firm: controls or anchors the platform.
- Niche firms: build products/services around it.
- Ecosystem health depends on standards, APIs, trust, incentives, and governance.

When to apply in a case: Use in platform competition, app economy, payment networks, or smart nation cases.

How to write in the exam: Identify the keystone platform and explain how other actors depend on it.

Simple example: Facebook, Apple App Store, Android, Grab, and smart city digital identity platforms all create ecosystems around shared digital infrastructure.

4 Unit 5: IT Infrastructure and Emerging Technologies

IT Infrastructure as a Service Platform

Core idea: IT infrastructure is the shared technology foundation and services that allow the whole enterprise to operate.

Components / steps:

- Hardware, software, networks, data management, facilities.
- Services: computing, telecom, data, application, IT management, standards, education, R&D.
- Service-platform view asks what business capabilities technology enables.

When to apply in a case: Use when a case asks whether infrastructure supports business growth, digital services, or scalability.

How to write in the exam: Connect infrastructure services to business capabilities such as mobile apps, analytics, cloud access, and customer service.

Simple example: Grab needs cloud, maps, mobile platforms, payment infrastructure, databases, and analytics to deliver fast, reliable digital services.

Technology Drivers of Infrastructure Evolution

Core idea: Infrastructure evolves because computing power, storage, networks, and standards improve over time.

Components / steps:

- Moore's law: processing power increases and cost falls.
- Law of mass digital storage: data volume rises while storage cost falls.
- Metcalfe's law: network value grows as users join.
- Declining communication costs and Internet standards.
- Acceptance of technology standards.

When to apply in a case: Use when explaining why cloud, mobile, big data, AI, or platform models became possible.

How to write in the exam: Show that business change is enabled by cheaper computing, cheaper data storage, better networks, and common standards.

Simple example: Big data analytics in Covid-19 response became feasible because data storage, network connectivity, and analytics platforms became affordable and scalable.

IT Infrastructure Components

Core idea: Enterprise infrastructure has multiple layers; weakness in one layer affects the whole digital business.

Components / steps:

- Computer hardware platforms.
- Operating system platforms.
- Enterprise software applications.
- Database management systems.
- Networking and telecommunications.
- Internet platforms.
- Consulting and system integration services.

When to apply in a case: Use when diagnosing what a company must build or upgrade for digital transformation.

How to write in the exam: List the relevant layers and explain the business role of each.

Simple example: A branchless bank needs mobile client devices, app servers, secure OS, databases, payment networks, Internet platforms, and integration with banking systems.

Cloud Computing: IaaS, PaaS, SaaS

Core idea: Cloud computing provides on-demand computing services over a network and helps firms scale without heavy upfront IT investment.

Components / steps:

- IaaS: infrastructure such as servers, storage, networking.
- PaaS: development platform, databases, middleware.
- SaaS: complete application delivered by vendor.
- Public, private, and hybrid cloud options.

- Risks: security, reliability, compliance, vendor dependence.

When to apply in a case: Use in cases involving cloud migration, scalability, cost reduction, or outsourced IT.

How to write in the exam: Explain why cloud fits the business need, then discuss risks and controls.

Simple example: Glory finding solutions in the cloud or Pick n Pay cloud migration can be framed as using cloud for faster deployment, lower capital cost, better BI access, and scalable analytics.

Virtualization, Edge, Green IT, and BYOD

Core idea: Modern infrastructure trends improve efficiency, flexibility, latency, sustainability, and user mobility.

Components / steps:

- Virtualization: one physical resource acts like many logical resources.
- Edge computing: processing near data source to reduce latency.
- Green IT: reduce energy consumption and environmental impact.
- BYOD/mobile platform: employees use mobile devices, requiring management and security.

When to apply in a case: Use when case has latency, mobile workforce, sustainability, data center cost, or device management issues.

How to write in the exam: Pick only the relevant trend and tie it to business outcome.

Simple example: A mobile timesheet app for Vinci Energies benefits from mobile platforms and cloud back-end; if field data is latency-sensitive, edge design may also help.

Total Cost of Ownership (TCO)

Core idea: TCO measures the full cost of owning technology, not just purchase price.

Components / steps:

- Hardware/software purchase or subscription.
- Installation and integration.
- Training and support.
- Maintenance, upgrades, downtime, security, administration.
- Switching and vendor lock-in costs.

When to apply in a case: Use when comparing cloud vs in-house, package vs custom, or new system investment.

How to write in the exam: Write that the cheaper purchase price may not be cheaper over time. Compare full lifecycle cost.

Simple example: A cloud database tool may reduce hardware cost for VF-Fiji, but the answer should also mention migration, training, security, subscriptions, and vendor dependence.

Competitive Forces Model for IT Infrastructure

Core idea: Infrastructure choices should reflect market demand, firm strategy, IT strategy, competitors, vendor power, and investment cost.

Components / steps:

- Market demand for services.
- Firm's business strategy.

- Firm's IT strategy, infrastructure, and cost.
- Competitor services.
- IT vendor assessment.
- Infrastructure investment level.

When to apply in a case: Use when a case asks how much or what kind of infrastructure a firm should build.

How to write in the exam: Avoid one-size-fits-all answers. Tie infrastructure level to competitive need.

Simple example: Deliveroo needs high uptime, maps, payments, and analytics because competitors can quickly win customers if the app is slow or unreliable.

5 Unit 6: Foundations of Business Intelligence, Databases, and Information Management

Data Hierarchy and File Environment Problems

Core idea: Good information management starts with organizing data and avoiding traditional file-system problems.

Components / steps:

- Hierarchy: database, file, record, field, entity, attribute.
- Traditional problems: redundancy, inconsistency, lack of flexibility, program-data dependence, poor security.
- DBMS solves these by centralizing and controlling data.

When to apply in a case: Use when a case has duplicated data, inconsistent reports, poor access, or manual spreadsheets.

How to write in the exam: Describe the current data problem, then recommend a database/DBMS approach.

Simple example: A customer service firm with separate customer files in sales, billing, and support will produce inconsistent service unless data is centralized.

DBMS and Logical vs Physical Views

Core idea: A DBMS separates how users see data from how data is physically stored, making data easier to manage and secure.

Components / steps:

- Centralizes related data.
- Controls redundancy and inconsistency.
- Separates logical and physical views.
- Supports multiple views for different users.
- Improves security and governance.

When to apply in a case: Use when explaining why a database is better than separate files.

How to write in the exam: Give examples of different views: payroll, benefits, insurance, sales, customer service.

Simple example: An employee database can support payroll, leave, benefits, and HR analytics without duplicating separate employee records.

Relational DBMS and SQL Operations

Core idea: Relational databases organize data as tables connected by keys, enabling flexible querying.

Components / steps:

- Table: rows and columns.
- Primary key: uniquely identifies records.
- Foreign key: links one table to another.
- SELECT: choose rows.
- JOIN: combine tables.
- PROJECT: choose columns.

When to apply in a case: Use when a case asks how data can be combined for reporting or analysis.

How to write in the exam: Explain the table relationships in simple business terms.

Simple example: For e-commerce, order table can join customer table and product table to show which customer segments buy which product categories.

Database Design: Normalization and ERD

Core idea: Database design structures data to reduce duplication and represent real-world entities and relationships.

Components / steps:

- Normalization: organize tables to reduce redundancy and update errors.
- Entity Relationship Diagram: maps entities, attributes, and relationships.
- Good design improves accuracy, flexibility, and reporting.

When to apply in a case: Use when a case involves poor data quality or building a new database.

How to write in the exam: Mention entities like customer, order, product, supplier, employee, branch, service ticket.

Simple example: In a delivery app, entities include customer, restaurant, driver, order, payment, location, and complaint.

Nonrelational, Cloud, Distributed Databases, and Blockchain

Core idea: Not all data fits traditional relational databases. Modern systems use flexible and distributed data models.

Components / steps:

- Nonrelational/NoSQL: flexible model for large distributed data sets.
- Cloud database: hosted database with pay-per-use pricing.
- Distributed database: stored across multiple physical locations.
- Blockchain: distributed ledger where records are verified and hard to alter.

When to apply in a case: Use when cases involve big data, global access, cloud DB tools, or secure transactions.

How to write in the exam: Choose the database type based on data volume, flexibility, distribution, and trust needs.

Simple example: Cloud DB tools helped VF-Fiji by improving scalability and access; blockchain is more relevant when transaction trust and auditability matter.

Big Data and the 3Vs

Core idea: Big data refers to massive data sets, often unstructured or semi-structured, that require new tools to manage and analyze.

Components / steps:

- Volume: huge quantity of data.
- Variety: structured, semi-structured, unstructured data.
- Velocity: speed of data generation and processing.
- Value comes from discovering patterns, anomalies, and relationships.

When to apply in a case: Use when cases involve sensors, social media, customer behavior, health data, sports data, or Covid-19 analytics.

How to write in the exam: State which V is most important in the case and what insight it creates.

Simple example: Big data in baseball uses high-volume performance and game data to identify player patterns and improve scouting or strategy.

Business Intelligence Infrastructure

Core idea: BI infrastructure consolidates and prepares data for analysis and decision making.

Components / steps:

- Data warehouse: current and historical data from core systems.
- Data mart: subset for a business unit or subject.
- Hadoop: distributed processing for big data.
- In-memory computing: fast analysis using RAM.
- Analytical platforms: optimized hardware/software for large datasets.

When to apply in a case: Use when a case asks how a firm turns data into dashboards, reports, or analytics.

How to write in the exam: Explain the data pipeline: operational systems to warehouse/mart to analytics tools to decision makers.

Simple example: Pick n Pay cloud migration for BI can be explained as building scalable infrastructure for consolidated reporting and faster analytics.

Analytical Tools: OLAP, Data Mining, Text Mining, Web Mining

Core idea: Analytical tools help discover relationships, patterns, trends, and predictions.

Components / steps:

- OLAP: multidimensional analysis by product, region, time, customer, etc.
- Data mining: associations, sequences, classification, clustering, forecasting.
- Text mining: extracts meaning from documents, messages, reviews.
- Web mining: analyzes web content, links, and user behavior.

When to apply in a case: Use when a case involves customer behavior, forecasting, segmentation, fraud, sentiment, or web/app activity.

How to write in the exam: Name the tool and the decision it supports.

Simple example: Astra can use data mining to classify customers, predict churn, and personalize service offers; text mining can analyze customer complaints.

6 Unit 7: Securing Information Systems

Security vs Control

Core idea: Security protects information systems from unauthorized access, alteration, theft, or damage. Controls ensure safety, accuracy, reliability, and compliance.

Components / steps:

- Security: policies, procedures, and technical measures.
- Controls: methods and organizational procedures to protect assets and ensure reliable records.
- Both are needed for trust, compliance, and continuity.

When to apply in a case: Use in cyberattack, fraud, data breach, or system reliability cases.

How to write in the exam: Differentiate prevention/protection from broader control and governance.

Simple example: Capital One data breach questions should cover both technical security weaknesses and control failures in cloud configuration, monitoring, and governance.

Vulnerability and Threat Analysis

Core idea: Information systems are vulnerable because networks are open, devices are mobile, software has errors, and users can be manipulated.

Components / steps:

- Internet, email, IM, P2P, Bluetooth, Wi-Fi.
- Malware: viruses, worms, Trojan horses, spyware, ransomware.
- Social engineering, phishing, spoofing.
- Insider threats and user errors.
- Hardware/software failures, disasters, loss of devices.

When to apply in a case: Use when a case asks why an attack happened or what risks exist.

How to write in the exam: Classify the threat source and show the business impact.

Simple example: Cyberattacks in APAC targeting the weakest firms can be explained through poor patching, phishing, weak access controls, and low security awareness.

Organizational Framework for Security and Control

Core idea: A firm needs a formal framework to manage risk, not just individual tools.

Components / steps:

- Information system controls.
- Risk assessment.
- Security policy.
- Disaster recovery planning and business continuity planning.
- Role of auditing.

When to apply in a case: Use as the main answer structure for any cybersecurity case.

How to write in the exam: Apply each component: identify asset, risk, policy, controls, recovery plan, audit.

Simple example: For PayPal digital resiliency, the answer should include controls, risk assessment, continuity planning, monitoring, and testing because payment platforms require trust and uptime.

Risk Assessment

Core idea: Risk assessment identifies valuable assets, likely threats, vulnerabilities, probability, impact, and controls.

Components / steps:

- Identify information assets and business processes.
- Identify threats and vulnerabilities.
- Estimate likelihood and business impact.
- Prioritize controls based on risk.
- Review continuously as threats change.

When to apply in a case: Use when asked what security measures should be adopted or how to prioritize cybersecurity investment.

How to write in the exam: Mention that not all risks can be eliminated; controls should match business impact.

Simple example: A bank should prioritize customer credentials, transaction systems, fraud monitoring, and regulatory data because impact is high.

Security Policy and Acceptable Use

Core idea: Security policy defines how people and systems should protect information.

Components / steps:

- Access rules and identity management.
- Acceptable use of systems, networks, email, mobile devices.
- Data classification and privacy rules.
- Incident response responsibilities.
- Training and enforcement.

When to apply in a case: Use when human behavior or unclear rules contributed to risk.

How to write in the exam: Write policy as a management control, not only a technical measure.

Simple example: If employees use personal devices for work, mobile policies and MDM are needed to control apps, encryption, updates, and remote wipe.

Tools and Technologies for Safeguarding Systems

Core idea: Security tools reduce risk by authenticating users, filtering traffic, detecting intrusions, encrypting data, and controlling devices.

Components / steps:

- Identity management and authentication: passwords, tokens, smart cards, biometrics, two-factor authentication.
- Firewalls: packet filtering, stateful inspection, NAT, application proxy.
- IDS/IPS, anti-malware, UTM.
- Wireless security: WPA2/WPA3.
- Encryption, SSL/TLS, digital certificates, PKI.
- Cloud and mobile controls: SLA, audits, MDM, encryption, segregation of corporate data.

When to apply in a case: Use when recommending controls for a cyber case.

How to write in the exam: Select controls that fit the threat: phishing needs training and MFA; malware needs patching and anti-malware; cloud risk needs configuration audits and SLAs.

Simple example: Capital One-style cloud risk requires IAM controls, least privilege, configuration monitoring, encryption, audit logs, and vendor/cloud responsibility clarity.

Disaster Recovery and Business Continuity

Core idea: Disaster recovery restores technology; business continuity keeps critical business processes running during disruption.

Components / steps:

- Disaster recovery plan: backup, recovery sites, restoration procedures.
- Business continuity plan: critical processes, roles, communication, alternate work methods.
- Testing and updating plans are essential.

When to apply in a case: Use for resiliency, outages, ransomware, payment disruptions, or natural disaster cases.

How to write in the exam: Differentiate system recovery from business process continuity.

Simple example: PayPal digital resiliency depends on redundant infrastructure and continuity processes so payments can continue even during incidents.

7 Unit 8: Enterprise Applications: ERP, SCM, CRM

Enterprise Resource Planning (ERP)

Core idea: ERP integrates key internal business processes using shared software modules and a common central database.

Components / steps:

- Modules: finance, HR, manufacturing, production, sales, marketing.
- Information entered in one process becomes available to others.
- Firms map business processes to software best practices.
- Customization should be controlled to avoid cost and complexity.

When to apply in a case: Use when a case has fragmented systems, duplicate data, slow reporting, or global process standardization.

How to write in the exam: Explain how ERP creates one version of the truth and supports firm-wide decisions.

Simple example: Versum ERP transformation can be framed as replacing fragmented processes with integrated enterprise data and standard workflows.

Business Value of Enterprise Systems

Core idea: Enterprise systems improve operational efficiency and management visibility.

Components / steps:

- Reduce redundant processes and systems.
- Improve process cycle time.
- Provide firm-wide information for decision making.

- Enable faster response to customers and suppliers.
- Include analytics for organizational performance.

When to apply in a case: Use when a question asks benefits of ERP or enterprise systems.

How to write in the exam: Link each benefit to measurable outcomes: cost, cycle time, accuracy, response speed.

Simple example: Alcoa used enterprise software to reduce redundant systems, centralize financial/procurement activities, and cut requisition-to-pay cycle time.

Supply Chain Management (SCM)

Core idea: SCM coordinates the network of organizations and processes for procuring materials, transforming them into products, and distributing products to customers.

Components / steps:

- Upstream supply chain: suppliers and sourcing.
- Internal supply chain: production and operations.
- Downstream supply chain: distribution and customers.
- SCM software: planning and execution systems.

When to apply in a case: Use in cases about inventory, logistics, delivery, supplier coordination, or demand-supply mismatch.

How to write in the exam: Identify upstream, internal, and downstream issues, then explain how information sharing improves flow.

Simple example: Lenzing balancing supply and demand is a classic SCM case: planning systems forecast demand while execution systems manage movement of products.

Bullwhip Effect

Core idea: Small changes in customer demand become amplified as information moves backward through the supply chain, causing excess inventory or shortages.

Components / steps:

- Demand signal distortion.
- Overordering by each participant “just in case.”
- Results: excess inventory, stockouts, higher costs, poor service.
- Reduced by real-time demand visibility, shared data, and better forecasting.

When to apply in a case: Use when cases mention inventory swings, demand uncertainty, Covid disruptions, or overstocking.

How to write in the exam: Explain the chain reaction and recommend shared demand data or pull systems.

Simple example: Corona disrupting supply chains increases uncertainty; firms need visibility and coordinated planning to avoid overreacting to demand spikes.

Push vs Pull Supply Chain

Core idea: Push systems build based on forecasts; pull systems respond to actual customer demand.

Components / steps:

- Push: build-to-stock, based on expected demand.

- Pull: demand-driven, customer orders trigger supply chain actions.
- Internet enables concurrent supply chains where suppliers adjust quickly.

When to apply in a case: Use when a case asks how digital systems change manufacturing or logistics.

How to write in the exam: State whether the firm should move from forecast-driven push to customer-driven pull.

Simple example: E-commerce delivery and food delivery platforms rely more on pull signals because customer orders trigger restaurant preparation, rider assignment, and delivery.

Customer Relationship Management (CRM)

Core idea: CRM captures, integrates, analyzes, and distributes customer data across all touchpoints to improve sales, service, marketing, and retention.

Components / steps:

- Single enterprise view of the customer.
- Sales force automation.
- Customer service and self-service.
- Marketing automation and cross-selling.
- Partner relationship management and employee relationship management where relevant.

When to apply in a case: Use for customer service, personalization, retention, churn, sales, and marketing cases.

How to write in the exam: Show how CRM connects customer data to revenue and satisfaction.

Simple example: CRM Helps Adidas can be answered through customer data integration, targeted marketing, loyalty, service improvement, and better customer lifetime value.

Operational vs Analytical CRM

Core idea: Operational CRM runs customer-facing processes; analytical CRM uses data warehouses and analytics to understand customers.

Components / steps:

- Operational CRM: sales force automation, call center, service support, marketing automation.
- Analytical CRM: OLAP, data mining, customer lifetime value, churn analysis, segmentation.
- Both together support better service and smarter decisions.

When to apply in a case: Use when a case asks what kind of CRM capability is needed.

How to write in the exam: Separate front-office execution from back-end analysis.

Simple example: Astra can use operational CRM for service requests and analytical CRM to identify high-value or at-risk customers.

Enterprise Application Challenges

Core idea: Enterprise applications create value but are difficult because they require business process, data, and organizational change.

Components / steps:

- High cost and long implementation time.
- Business process redesign.

- Organizational learning and resistance.
- Switching costs and vendor dependence.
- Data standardization, cleansing, and governance.

When to apply in a case: Use when a case asks risks or implementation issues of ERP/CRM/SCM.

How to write in the exam: Mention that the system may fail if the firm ignores process change and data quality.

Simple example: Versum ERP or Lenzing SCM projects require clean data, process ownership, training, and governance, not just software installation.

8 Unit 9: E-Commerce, Digital Markets, and Digital Goods

Unique Features of E-Commerce Technology

Core idea: E-commerce differs from traditional markets because Internet technologies change reach, cost, information, and interaction.

Components / steps:

- Ubiquity: available anytime/anywhere.
- Global reach: crosses geographic boundaries.
- Universal standards: common Internet standards.
- Richness: video, audio, text.
- Interactivity: two-way communication.
- Information density: transparency and lower information cost.
- Personalization/customization.
- Social technology: user-generated content and networks.

When to apply in a case: Use when explaining why an online business model works or why digital competition is powerful.

How to write in the exam: Pick the features visible in the case rather than listing all eight mechanically.

Simple example: Deliveroo uses ubiquity, mobile reach, interactivity, personalization, and information density to connect customers, restaurants, and riders.

Digital Markets Concepts

Core idea: Internet markets reduce information asymmetry and transaction costs, while enabling new pricing and channel structures.

Components / steps:

- Lower search costs and transaction costs.
- Lower menu costs.
- Dynamic pricing and price discrimination.
- Reduced information asymmetry.
- Disintermediation and reintermediation.
- Switching costs and delayed gratification.

When to apply in a case: Use in marketplace, online retail, pricing, or platform questions.

How to write in the exam: Explain how digital markets change buyer/seller power and channel economics.

Simple example: iDEAL in the EU market can be linked to lower transaction friction and trust in digital payments.

Types of E-Commerce

Core idea: E-commerce can be classified by participants and platform.

Components / steps:

- B2C: business to consumer, e.g., Amazon, Flipkart.
- B2B: business to business, e.g., Indiamart.
- C2C: consumer to consumer, e.g., OLX.
- M-commerce: mobile commerce through smartphones and apps.

When to apply in a case: Use when identifying what kind of e-commerce model a case represents.

How to write in the exam: Mention both participant model and platform if relevant.

Simple example: Deliveroo is mainly B2C/m-commerce; OLX is C2C; a procurement marketplace is B2B.

E-Commerce Business Models

Core idea: Digital businesses create value through different model types.

Components / steps:

- Portal: gateway to content/services.
- E-tailer: online retailer.
- Content provider: digital content.
- Transaction broker: processes transactions.
- Market creator: digital marketplace.
- Service provider: online service.
- Community provider: user community/social network.

When to apply in a case: Use when a case asks what business model a digital firm uses.

How to write in the exam: Identify the model and explain how it creates customer value.

Simple example: Deliveroo is a market creator and service provider connecting customers, restaurants, and riders.

E-Commerce Revenue Models

Core idea: Revenue model explains how the digital business earns money.

Components / steps:

- Advertising.
- Sales.
- Subscription.
- Free/freemium.
- Transaction fee.

- Affiliate.

When to apply in a case: Use when asked how an e-commerce firm monetizes.

How to write in the exam: Separate business model from revenue model. A firm can be a marketplace and earn transaction fees.

Simple example: A food delivery platform may earn restaurant commissions, delivery fees, advertising placements, and subscription loyalty revenue.

Social, Mobile, and Local Commerce

Core idea: Modern e-commerce is increasingly driven by smartphones, social interactions, and location-based services.

Components / steps:

- Social commerce: recommendations, reviews, influencers, social ads.
- Mobile commerce: app-based transactions, payments, notifications.
- Local commerce: location-based offers and delivery.
- Conversational commerce: chatbots, messaging, and service conversations.

When to apply in a case: Use in cases involving mobile apps, social engagement, food delivery, or customer engagement.

How to write in the exam: Connect the technology to customer convenience and marketing effectiveness.

Simple example: Engaging Socially can be answered through social CRM, user-generated content, brand conversations, and targeted social campaigns.

9 Unit 10: Managing Knowledge and Artificial Intelligence

Knowledge Dimensions

Core idea: Knowledge is an important firm asset, but it has different forms and must be managed intentionally.

Components / steps:

- Data, information, knowledge, wisdom.
- Tacit knowledge: personal, experience-based, hard to codify.
- Explicit knowledge: documented, structured, shareable.
- Knowledge is situational and located in people, documents, databases, routines, and teams.
- Knowledge-based core competencies can create competitive advantage.

When to apply in a case: Use when a case involves expertise, learning, best practices, training, or knowledge loss.

How to write in the exam: Explain what knowledge exists and how the firm should capture or apply it.

Simple example: Predictive maintenance in oil and gas depends on explicit sensor data plus tacit engineering knowledge about equipment behavior.

Knowledge Management Value Chain

Core idea: KM creates value by moving knowledge through acquisition, storage, dissemination, and application.

Components / steps:

- Acquisition: capture tacit/explicit knowledge, documents, data, expert networks.
- Storage: databases, document systems, repositories.
- Dissemination: portals, wikis, search, collaboration tools, training.
- Application: new processes, products, services, and decisions.

When to apply in a case: Use as the main structure for knowledge management cases.

How to write in the exam: Apply all four stages and show where the firm is weak.

Simple example: Eastspring enterprise system building can be framed as acquiring project and business knowledge, storing it centrally, sharing it across teams, and applying it to better enterprise processes.

Types of Knowledge Management Systems

Core idea: Different KM systems support different knowledge needs.

Components / steps:

- Enterprise-wide KMS: firm-wide collection, storage, distribution, and application of digital content and knowledge.
- Knowledge work systems: specialized tools for engineers, scientists, designers, analysts.
- Intelligent techniques: expert systems, machine learning, data mining, neural networks, agents.

When to apply in a case: Use when asked what system should support knowledge work or expertise.

How to write in the exam: Match the system type to the user and knowledge problem.

Simple example: CAD and AR systems support designers/engineers; enterprise content management supports document sharing across the firm.

Enterprise Content Management and LMS

Core idea: ECM manages documents and semi-structured knowledge; LMS manages learning delivery and tracking.

Components / steps:

- ECM: capture, store, retrieve, distribute, preserve documents and digital assets.
- Taxonomy: classification system for finding knowledge.
- LMS: deliver, track, assess training and learning.
- MOOCs and web-based learning can scale knowledge distribution.

When to apply in a case: Use for document-heavy, training-heavy, or compliance-heavy cases.

How to write in the exam: Mention taxonomy if the problem is finding the right knowledge.

Simple example: A hospital or university digitization project needs ECM for policies/documents and LMS for training staff or students.

Expert Systems

Core idea: Expert systems capture human expertise as IF-THEN rules to make recommendations in a narrow domain.

Components / steps:

- Knowledge base: expert rules.
- Inference engine: applies rules to facts.

- Best for structured expert reasoning with clear rules.
- Limitation: less flexible when environments change or tacit judgment is needed.

When to apply in a case: Use when a case involves codifying expert decisions, diagnosis, compliance checks, or troubleshooting.

How to write in the exam: Show what rules would be captured and where human oversight remains needed.

Simple example: A medical decision support system can encode diagnostic rules, but doctors still evaluate context and responsibility.

Machine Learning Types

Core idea: ML recognizes patterns in large datasets instead of relying only on fixed rules.

Components / steps:

- Supervised learning: labeled data; classification and regression.
- Unsupervised learning: unlabeled data; clustering, associations, anomaly detection.
- Reinforcement learning: trial-and-error to maximize reward; sequential decisions.

When to apply in a case: Use in AI, prediction, fraud, recommendation, autonomous systems, or customer analytics cases.

How to write in the exam: Choose the ML type based on data and task.

Simple example: AI beats radiologists uses supervised learning on labeled medical images; fraud detection can use anomaly detection; self-driving cars involve reinforcement learning and computer vision.

Neural Networks, NLP, Computer Vision, Robotics, and Intelligent Agents

Core idea: Advanced AI techniques automate pattern recognition, language, visual understanding, physical action, and background decision tasks.

Components / steps:

- Neural networks: pattern detection from large data.
- NLP: understand and analyze human language.
- Computer vision: interpret images/video.
- Robotics: machines controlled by software and sensors.
- Intelligent agents: background software acting on behalf of users/processes.

When to apply in a case: Use when the case is specifically about AI applications.

How to write in the exam: Define the technique and state the business use, risk, and control needed.

Simple example: Who uses your face is a computer vision/privacy case; self-driving cars combine computer vision, sensors, ML, and robotics.

10 Unit 11: Enhancing Decision Making

Types of Decisions

Core idea: Decision support depends on whether the decision is structured, semi-structured, or unstructured.

Components / steps:

- Structured: routine, repetitive, clear procedure.
- Semi-structured: part routine, part judgment.
- Unstructured: requires judgment, evaluation, insight.
- Operational managers handle more structured decisions; senior executives handle more unstructured decisions.

When to apply in a case: Use when asked what kind of decision a case presents or what system supports it.

How to write in the exam: Classify the decision and match it to TPS/MIS/DSS/ESS or AI automation.

Simple example: Overtime pay is structured; pricing strategy is semi-structured; entering a new digital market is unstructured.

Simon's Decision-Making Process

Core idea: Decision making follows four stages: intelligence, design, choice, and implementation.

Components / steps:

- Intelligence: discover and understand the problem.
- Design: identify and explore solutions.
- Choice: choose among alternatives.
- Implementation: make the chosen solution work and monitor results.

When to apply in a case: Use in almost any case asking how managers should decide.

How to write in the exam: Structure the answer by the four stages and include data needed at each stage.

Simple example: For predictive maintenance, intelligence identifies failure patterns, design compares maintenance options, choice selects schedule/model, implementation monitors equipment outcomes.

Managerial Roles and IS Support

Core idea: Managers perform interpersonal, informational, and decisional roles, and IS can support some but not all of them.

Components / steps:

- Interpersonal: figurehead, leader, liaison.
- Informational: nerve center, disseminator, spokesperson.
- Decisional: entrepreneur, disturbance handler, resource allocator, negotiator.
- IS supports information and analysis but managerial judgment remains important.

When to apply in a case: Use when asked whether systems replace managers or how managers use information.

How to write in the exam: Emphasize that IS assists decision making but cannot remove leadership, politics, ethics, or judgment.

Simple example: A BI dashboard can support a city manager's nerve-center role, but negotiations and public accountability still require human leadership.

Real-World Decision-Making Constraints

Core idea: IT does not automatically produce good decisions because information quality, management filters, and organizational politics interfere.

Components / steps:

- Information quality: accuracy, timeliness, completeness, consistency.
- Management filters: bias and selective attention.
- Organizational inertia and politics: resistance to decisions requiring change.

When to apply in a case: Use when a case has failed analytics, ignored dashboards, or bad decisions despite data.

How to write in the exam: Write that decision support must be paired with data governance and change management.

Simple example: An algorithm recommending process change may be rejected if managers distrust the data or fear loss of power.

High-Velocity Automated Decision Making

Core idea: Some highly structured decisions are automated through algorithms, reducing or removing human intervention.

Components / steps:

- Software captures intelligence, design, choice, and implementation.
- Useful for repetitive, high-speed, rule-based decisions.
- Requires safeguards, monitoring, and regulation.

When to apply in a case: Use in algorithmic trading, fraud detection, automated credit approval, or platform matching cases.

How to write in the exam: Mention benefits and risks: speed and scale vs errors, bias, opacity, and loss of human oversight.

Simple example: Should an algorithm make our decision is directly about whether automated decisions need transparency, safeguards, and human accountability.

Business Intelligence Environment

Core idea: BI environment combines data, infrastructure, analytics, users, delivery platforms, and interfaces for decision support.

Components / steps:

- Data from business environment.
- BI infrastructure.
- Business analytics toolset.
- Managerial users and methods.
- Delivery platform: MIS, DSS, ESS.
- User interface: dashboards and data visualization.

When to apply in a case: Use when describing how a firm should build decision support capability.

How to write in the exam: Explain the full environment, not just dashboard screens.

Simple example: A cloud BI migration needs clean source data, warehouse infrastructure, analytics tools, trained managers, and clear dashboards.

BI and Analytics Capabilities

Core idea: BI systems deliver information through reports, dashboards, queries, drill-down, forecasts, and scenario analysis.

Components / steps:

- Production reports.
- Parameterized reports and pivot tables.
- Dashboards and scorecards.
- Ad hoc query/search/report creation.
- Drill-down.
- Forecasts, scenarios, and what-if analysis.

When to apply in a case: Use when asked what BI tools managers need.

How to write in the exam: Map capability to decision need.

Simple example: An e-commerce dashboard should show daily orders, allow drill-down by city/product, and support what-if analysis for pricing or delivery capacity.

Predictive, Big Data, Operational, and Location Analytics

Core idea: Different analytics types support different decision problems.

Components / steps:

- Predictive analytics: estimates probability of future events using historical data.
- Big data analytics: analyzes massive customer, social, sensor, or public data.
- Operational intelligence: real-time monitoring of business activity/IoT data.
- Location analytics/GIS: maps spatial patterns for decisions.

When to apply in a case: Use when the case mentions prediction, smart cities, sensors, maps, or real-time monitoring.

How to write in the exam: Choose the analytics type and explain the decision it improves.

Simple example: GIS helps Land O'Lakes manage assets by visualizing location-based assets and operational patterns; Singapore smart city cases use location and sensor analytics.

Balanced Scorecard and Business Performance Management

Core idea: Senior managers use balanced scorecards and KPIs to translate strategy into measurable performance.

Components / steps:

- Financial perspective.
- Business process perspective.
- Customer perspective.
- Learning and growth perspective.
- KPIs track progress toward strategic targets.

When to apply in a case: Use when a case asks how executives should measure success of an IS initiative.

How to write in the exam: Give KPIs for each dimension, not only financial metrics.

Simple example: For a digital transformation project: financial KPI is cost reduction, process KPI is cycle time, customer KPI is satisfaction, learning KPI is employee adoption/training.

11 Unit 12: Building Information Systems

Systems Development and Organizational Change

Core idea: Building a new system is planned organizational change. Different levels of change carry different risk and reward.

Components / steps:

- Automation: replace manual tasks, improve efficiency.
- Rationalization: streamline standard operating procedures.
- Business process redesign: analyze, simplify, redesign workflows.
- Paradigm shift: rethink business model and nature of organization.

When to apply in a case: Use when a case asks what kind of organizational change a system creates.

How to write in the exam: Classify the change level and explain risk/reward.

Simple example: A mobile timesheet app may automate reporting; DP World business process redesign may restructure workflows; a branchless bank is closer to paradigm shift.

Business Process Management (BPM) Steps

Core idea: BPM provides tools and methods to analyze, design, optimize, and continuously improve processes.

Components / steps:

1. Identify processes for change.
2. Analyze existing processes.
3. Design the new process.
4. Implement the new process.
5. Continuously measure performance.

When to apply in a case: Use for process redesign cases.

How to write in the exam: Apply each step to the case and include performance indicators.

Simple example: DP World process redesign should be answered by identifying bottleneck port/logistics processes, analyzing current flow, designing digital workflows, implementing, and measuring turnaround time/cost.

Core Activities in Systems Development

Core idea: System development turns an organizational problem or opportunity into an information system solution.

Components / steps:

- Systems analysis.
- Systems design.
- Programming.
- Testing.

- Conversion.
- Production and maintenance.

When to apply in a case: Use when asked how to build or implement a system.

How to write in the exam: Explain each activity briefly and connect it to user requirements and business goals.

Simple example: For Vinci Energies mobile timesheet app, analysis identifies field-worker needs, design specifies app/workflow, programming builds it, testing validates entries, conversion moves users, maintenance improves it.

Systems Analysis and Design

Core idea: Analysis defines the problem and information requirements; design specifies how the system will meet them.

Components / steps:

- Analysis: problem, causes, solution options, information requirements, feasibility, system proposal.
- Design: system specifications, managerial/organizational/technical components.
- End users must be involved because poor requirements cause failure.

When to apply in a case: Use when a case has unclear requirements, failed system, or new system proposal.

How to write in the exam: State information requirements and user involvement clearly.

Simple example: A timesheet app must collect project, employee, date, hours, approval status, and exceptions; managers and field employees must shape the design.

Testing and Conversion Strategies

Core idea: Testing ensures the system works; conversion moves from old system to new system.

Components / steps:

- Testing: test plan, system testing, acceptance testing.
- Parallel conversion: old and new run together.
- Direct cutover: switch immediately.
- Pilot: try in one part of organization.
- Phased: implement in stages.

When to apply in a case: Use when asked how to implement safely.

How to write in the exam: Choose strategy based on risk, urgency, size, and criticality.

Simple example: For payroll or banking, parallel/pilot is safer; for a small low-risk internal tool, direct cutover may be acceptable.

Structured vs Object-Oriented Methodologies

Core idea: Different modeling approaches represent systems differently.

Components / steps:

- Structured methodology: top-down, process-oriented, separates data from process, often uses DFD.
- Object-oriented development: combines data and processes in objects, uses classes and inheritance, more iterative and reusable.

When to apply in a case: Use when asked about systems modeling/design methodologies.

How to write in the exam: Choose structured for stable process flows; object-oriented for reusable, complex interactive systems.

Simple example: A university registration DFD suits structured modeling; a mobile app with reusable objects like user, timesheet, project, approval suits object-oriented design.

Traditional SDLC / Waterfall

Core idea: Traditional systems life cycle divides development into formal sequential stages.

Components / steps:

- One stage finishes before the next begins.
- Formal specifications and paperwork.
- Clear division between end users and IS specialists.
- Useful for large, complex, stable requirements.
- Weakness: costly, slow, inflexible.

When to apply in a case: Use when a case needs strong control, compliance, or stable requirements.

How to write in the exam: Mention why waterfall may or may not fit the case.

Simple example: A core banking compliance system may need formal SDLC controls, but a customer mobile feature may need agile speed.

Prototyping

Core idea: Prototyping rapidly builds an experimental system so users can evaluate and refine requirements.

Components / steps:

1. Identify user requirements.
2. Develop initial prototype.
3. Use prototype.
4. Revise and enhance prototype.

When to apply in a case: Use when requirements are uncertain, user interface matters, or users need to visualize the system.

How to write in the exam: State advantages and risks: better user fit, but may skip documentation, testing, and scalability.

Simple example: For a mobile timesheet app, prototyping lets field employees test screens and approval flows before final development.

Packages, SaaS, Outsourcing, and RFP

Core idea: Firms do not always build from scratch. They can buy packages, subscribe to cloud software, or outsource development.

Components / steps:

- Packages/SaaS save time and money but may require process adaptation.
- RFP compares vendors on function, flexibility, usability, resources, database, maintenance, documentation, vendor quality, and cost.
- Outsourcing adds flexibility and skills but creates hidden costs and dependency.

When to apply in a case: Use when a case asks build vs buy vs outsource.

How to write in the exam: Compare business fit, cost, speed, risk, vendor dependence, and control.

Simple example: A small firm needing CRM may choose SaaS; a firm with unique strategic process may custom-build or heavily configure.

RAD, JAD, Agile, Automated Testing, DevOps, Low-Code/No-Code

Core idea: Digital firms use faster development methods to deliver systems quickly and iteratively.

Components / steps:

- RAD: create workable systems quickly.
- JAD: users and specialists jointly define requirements/design.
- Agile: break work into short iterations with working software.
- Automated testing: repeatable tests compare expected and actual results.
- DevOps: combines development and operations for continuous delivery.
- Low-code/no-code: visual tools reduce hand-coding, but require governance.

When to apply in a case: Use when a case emphasizes speed, mobile apps, changing requirements, or software quality.

How to write in the exam: Choose method based on uncertainty, speed, control, and risk.

Simple example: McAfee automated software testing is a direct example: automated tests speed quality assurance and reduce errors across repeated software releases.

Case Themes and Framework Hooks

Uploaded case theme	Likely framework hook	Exam angle
Changes in Financial Industry	Digital firm, strategic objectives, Porter forces	Explain fintech disruption, branchless service, customer expectations, survival pressure.
Jurong Health Services digital transformation	Sociotechnical systems, IT governance, BPM, security	Healthcare systems need workflow redesign, data quality, privacy, and user adoption.
City of Mississauga goes digital	E-government, business process, BI/location analytics	Citizen services become faster, transparent, and data-driven.
Virtual Education at Ahlia University	Collaboration, social business, LMS, sociotechnical	Platform plus faculty/student process change.
N26: bank without branches	Digital business model, competitive strategy, cloud/mobile, security	Low-cost branchless model, app-based intimacy, trust and compliance.
Shipping Wars	Porter forces, value chain, SCM	Compete through logistics visibility, speed, tracking, customer service.
Singapore as Smart Nation	Value web, ecosystem, BI, location analytics, e-government	Public value through interconnected agencies, citizens, sensors, and platforms.
Grab leverages IT	Network strategy, IT infrastructure, CRM, analytics	Platform value grows with users/drivers/partners and data-driven services.
Glory cloud solutions	Cloud computing, TCO, infrastructure strategy	Scalability and cost flexibility balanced with vendor/security risk.

Open source innovation	Software platforms, standards, ecosystem	Community innovation, lower cost, flexibility, support/governance risks.
Astra customer-driven service	BI, data mining, analytical CRM	Use customer data for segmentation, churn prediction, personalization.
Cloud DB tools helped VF-Fiji	Cloud database, TCO, BI infrastructure	Better access, scalability, cost model, and data management.
Big Data baseball	Big data 3Vs, data mining, predictive analytics	Use performance data to improve scouting, training, and game decisions.
Big Data in APAC Covid-19	Big data, operational/location analytics	Use large, fast, varied data for public health response.
Cyberattacks in APAC	Security framework, risk assessment, policy	Weakest controls are exploited; need layered controls and governance.
Capital One	Cloud security, IAM, audit, risk assessment	Cloud responsibility, misconfiguration, monitoring, and data breach impact.
PayPal digital resiliency	DR/BCP, controls, operational risk	Payment platforms require high availability, monitoring, redundancy.
Bulgaria cyber/privacy case	Security policy, legal/regulatory controls	Public data and national systems need strong governance and accountability.
Lenzing supply-demand	SCM, bullwhip, push/pull, BI	Align demand planning, inventory, sustainability, and supply visibility.
CRM Helps Adidas	CRM, analytical CRM, social CRM	Customer intimacy, segmentation, loyalty, CLTV, churn reduction.
Versum ERP transformation	ERP, enterprise challenges, process change	Integrated data and standard processes, with change management risk.
Corona disrupts supply chain	Bullwhip, SCM, global supply chain risk	Need resilience, visibility, safety stock, and demand-driven planning.
E-commerce dashboard	BI capabilities, MIS/DSS/ESS, e-commerce	Dashboards, drill-down, KPIs, sales and operations decisions.
Deliveroo food delivery	E-commerce models, m-commerce, network effects, SCM	Marketplace plus transaction fees, mobile/local commerce, delivery coordination.
Engaging socially	Social business, social CRM, e-commerce marketing	Conversations, user content, targeted campaigns, brand trust.
iDEAL EU market	Digital markets, payment systems, transaction cost	Reduce payment friction and build trust in single digital market.
AI beats radiologists	Supervised learning, computer vision, ethics	Accuracy, data training, human oversight, responsibility.
Who uses your face	Computer vision, privacy, security policy	Facial recognition needs consent, governance, security, and fairness.
Will AI kill jobs	AI, sociotechnical systems, organizational change	Automation changes work; reskilling and process redesign matter.
Self-driving cars	ML, computer vision, robotics, automated decision making	Safety, regulation, edge cases, human accountability.
Eastspring enterprise system	KMS, enterprise systems, system development	Targeted enterprise system building requires knowledge capture and integration.

Predictive maintenance oil and gas	Predictive analytics, IoT, BI, DSS	Sensors and models reduce downtime and maintenance cost.
GIS helps Land O'Lakes	GIS/location analytics, decision support	Spatial data improves asset management and field decisions.
Should algorithm decide	Automated decisions, ethics, decision quality	Bias, transparency, safeguards, and human review.
Vinci mobile timesheet app	Prototyping, mobile app development, BPM	Automate field reporting and approvals with user-centered design.
DP World process redesign	BPM, BPR, systems development	Redesign workflows before automating them.
McAfee automated testing	Automated testing, DevOps, agile	Quality and release speed improve through repeatable test automation.
Pick n Pay cloud BI	Cloud migration, BI infrastructure, dashboards	Cloud supports scalable reporting and decision making.

Last-Minute Exam Writing Phrases

- "The case is not only a technology issue; it also requires organizational process change and managerial governance."
- "The main business value of the system is visible in reduced cycle time, better information quality, and faster decision making."
- "Using the framework, the company should first identify the process/data risk, then redesign the workflow, then implement controls and measurement."
- "The firm should avoid technology-first implementation. It should begin with information requirements and business objectives."
- "The success of this system depends on data quality, user adoption, integration with existing systems, and continuous monitoring."
- "The recommended approach is to combine operational systems for execution with analytical systems for decision support."
- "For a case-based answer, the strongest point is the link between the case evidence and the business impact, not the definition alone."